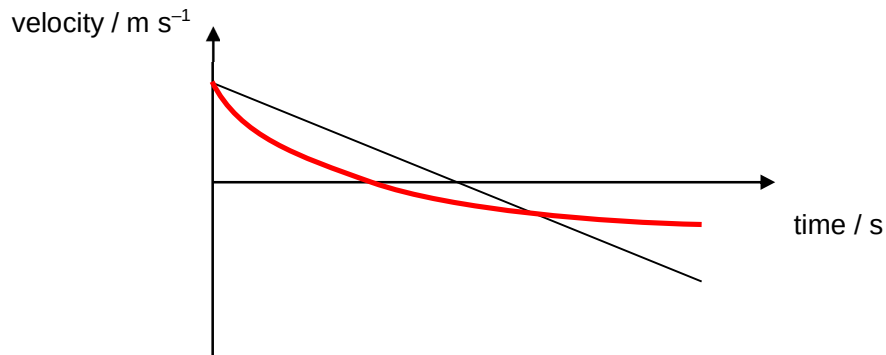


4 (a)



curve from $+15 \text{ m s}^{-1}$ steepest at first then gentler and gentler [B1]
gradient at $v = 0$ should be same as that of original line. [B1]
areas under graph above and below x-axis are similar. [B1]

(b) greater resultant (downward opposing) force, so lesser height [A1]

5 (a) No resultant (external) force (acts on the car-truck system) so (by principle of conservation